SNAPDRAGON® AR1 GEN 1 PLATFORM

The Snapdragon AR1 Gen 1 Platform is designed to power next-generation smart glasses, delivering seamless capture, livestreaming, notifications, and powerful on-glass AI.

Effortless photo and video capture

Snapdragon AR1 Gen 1 premium dual ISPs enable amazing 12 MP photo and 6 MP video capture—directly from your glasses. Stay in the moment to capture memorable visuals that WOW with vivid colors and clarity.

Support for binocular displays

Snapdragon AR1 Gen 1 revolutionizes the smart glass experience. With support for single or double lens displays, 3DoF, and 1280 x 1280 resolution per eye @ 60 FPS, notifications and alerts blend seamlessly into the user's field of view with a sense of depth that enhances visual comfort.

Next-gen AI capabilities

Powered by the 3rd Generation Qualcomm® Hexagon™ NPU and the AR-grade Engine for Visual Analytics, Snapdragon AR1 Gen 1 enhances image and audio quality to ensure life-like memory capture. Plus, visual search and real-time translation capabilities enable the ultimate personal assistant experience.

Seamless sharing and streaming

Thrilling experiences are better shared. The blazing-fast Qualcomm® FastConnect™ 7800 Mobile Connectivity System enables Wi-Fi 7 and is qualified against Bluetooth® Core 5.4 wireless technology, along with increased memory for seamless, faster livestreaming directly from the glasses*.

Low-power audio capabilities

Eight microphones and AI-powered targeted capture enable crystal-clear audio and video capture, while Qualcomm® Noise and Echo Cancellation allows clearer calls.

Optimized thermals and power

Snapdragon AR1 Gen 1 is optimized for lightweight, battery-efficient smart glasses, blending smart and style with an advanced process node and an optimized package.

* With support for the latest FastConnect Suite, users will experience lag-free and shared XR experiences with ultra-fast Wi-Fi 7 and Wi-Fi 6E and Bluetooth—qualified against Bluetooth Core 5.4—wireless connectivity for 60% higher throughput and 50% lower latency compared to previous FastConnect generations.

Materials subject to change without notice.
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SPECIFICATIONS & FEATURES

CPU

Qualcomm® Kryo™ CPU

4x cores 1.9 GHz

64-bit Architecture

GPU

Qualcomm® Adreno™ GPU

Support for power-efficient sparse AR content generation

Support for OpenGL ES 3.2 & Vulkan 1.1

Display

Binocular Display

1280 x 1280p 60 Hz RGB display output

Video

Decode: 1920 x 1080 30 FPS

Encode: 6 MP 30 FPS

Concurrency: 720p30 FPS encode + 720p30 FPS decode

Memory

Support for LPDDR4x memory, up to 2100 MHz

Camera

Qualcomm Spectra™ Image Signal Processor

- Dual 14-bit ISPs
- Up to 12 MP computational HDR and portrait mode photo capture
- Advanced auto-exposure, and auto white balance
- Hardware Electronic Image Stabilization
- Up to 6 MP 30 FPS video capture
- Slow-mo video capture at 720p120 FPS
- AI Face Detection and Facial Landmark Detection

Image quality and imaging improvement

- Mega Low-Light Photo Capture
- Motion Compensated Temporal Filtering
- Smart ASF for better alignment and halo control
- LCAC for wide-angle chromatic aberration correction

AI

Qualcomm® AI Engine

- Hexagon NPU
- Hexagon scalar, vector, and tensor accelerators
- Support for precisions (INT4, INT8)

Qualcomm® Sensing Hub

- Micro NPU for audio and sensors
- Support for INT8 precision

Audio

Qualcomm Aqstic™ Speaker Max for stereo speakers

Speaker processing and speaker protection

Snapdragon Voice Activation

Low-power audio with Bluetooth A2DP and Hands-Free Profile

Qualcomm Noise and Echo Cancellation

Qualcomm® Audio and Voice Communication Suite

Always-on audio

Stereo speakers

Up to eight microphones

Wi-Fi & Bluetooth® technology

FastConnect 7800 Mobile Connectivity System

- Wi-Fi Generation: Wi-Fi 7, Wi-Fi 6
- Peak speed: 5.8 Gbps
- 802.11be, 803.11ax, 802.11ac, 802.11a/b/g/n
- Wi-Fi Security: WPA3-Enterprise, WPA3-Enhanced Open, WPA3 Easy Connect, WPA3-Personal
- Qualified against Bluetooth Core 5.4

Security

Trust Management Engine (Root of Trust), along with platform-level security foundations

Qualcomm® Trusted Execution Environment (TEE) for use cases requiring higher processing assurance

To learn more visit: qualcomm.com

Snapdragon

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